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Intelligent Workflow Automation Agent

Production-ready workflow automation platform
with visual editor, AI-powered steps, and multi-trigger support

TypeScript | React | Express | Prisma | Vite

v1.0.0 | 2026

The Problem

! Manual Processes

Repetitive tasks consume valuable engineering time

! No Visibility

Hard to track workflow execution and failures

! Scattered Tools

Multiple disconnected tools for different automation needs

! No Code Execution

Lack of safe, sandboxed code execution in workflows

Goal: Build a unified, intelligent workflow automation platform that empowers teams to design, execute, and monitor complex workflows with ease.

Our Solution

A full-stack workflow automation platform with:

- + Visual drag-and-drop workflow editor (React Flow)
- + Multiple node types: AI, HTTP, Code, Condition, Email, etc.
- + Multi-trigger support: Manual, Schedule (Cron), Webhook, Event
- + Real-time execution monitoring via WebSocket
- + Production-ready with Prisma ORM, BullMQ, and Redis
- + Role-based audit logging for compliance

Architecture

FRONTEND	BACKEND	INFRASTRUCTURE
React 18	Express.js	SQLite / PostgreSQL
Vite	TypeScript	Redis (BullMQ)
React Flow	Prisma ORM	WebSocket
React Router	Socket.IO	Cron Scheduler

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Data Flow:

1. User designs workflow in Visual Editor (React Flow)
2. Workflow saved to database via REST API
3. Trigger activated (Manual / Schedule / Webhook / Event)
4. Workflow Engine executes nodes sequentially
5. Real-time status updates via WebSocket
6. Results stored, audit log updated

Key Features

Visual Editor	Drag-and-drop workflow design with React Flow canvas
AI Integration	OpenAI/Anthropic powered intelligent workflow steps
Multi-Trigger	Manual, Cron schedule, Webhook, and Event-based triggers
Code Execution	Sandboxed JavaScript/Python code execution in workflows
Real-time	Live execution monitoring via WebSocket connections
Audit Trail	Complete execution history with compliance logging

Node Types

13 built-in node types for building complex workflows:

Node	Description	Use Case
Start / End	Entry and exit points	Workflow boundaries
AI Step	OpenAI/Anthropic integration	Data analysis, content generation
HTTP Request	REST API calls	External service integration
Code	JS/Python execution	Custom logic, data processing
Condition	Conditional branching	If/else logic paths
Delay	Time-based waiting	Scheduling, rate limiting
Transform	Data transformation	JSON manipulation, mapping
Email	Send email notifications	Alerts, reports
Notification	Push notifications	Real-time alerts
Parallel	Fork execution	Concurrent processing
Merge	Join branches	Result aggregation
Webhook	External trigger	Event-driven workflows

Tech Stack

CATEGORY	TECHNOLOGY	PURPOSE
Runtime	Node.js 20+	Server runtime environment
Language	TypeScript	Type-safe development
Backend	Express.js	REST API framework
ORM	Prisma	Database access layer
Database	SQLite / PostgreSQL	Data persistence
Queue	BullMQ + Redis	Job queue management
Frontend	React 18	UI framework
Bundler	Vite	Build tool and dev server
Canvas	React Flow	Visual workflow editor
Real-time	Socket.IO	WebSocket communication
Scheduler	node-cron	Cron-based scheduling
AI	OpenAI SDK	AI model integration
Validation	Zod	Schema validation
Logging	Pino	Structured logging

API Endpoints

RESTful API with full CRUD operations:

METHOD	ENDPOINT	DESCRIPTION
GET	/api/health	Health check
GET	/api/workflows	List all workflows
POST	/api/workflows	Create workflow
GET	/api/workflows/:id	Get workflow detail
PUT	/api/workflows/:id	Update workflow
DELETE	/api/workflows/:id	Delete workflow
POST	/api/workflows/:id/run	Execute workflow
GET	/api/workflows/:id/runs	Get workflow runs
POST	/api/webhooks/:id	Webhook trigger
GET	/api/schedules	List schedules
GET	/api/events	List events

Database Schema

Prisma ORM with SQLite (dev) / PostgreSQL (prod):

Workflow

id, name, description, status

triggerType (manual/schedule/webhook/event)

nodes (JSON), edges (JSON), variables (JSON)

version, createdAt, updatedAt

WorkflowRun

id, workflowId (FK), status

input (JSON), output (JSON), error

nodeStates (JSON), startedAt, completedAt, duration

Schedule

id, workflowId (FK), cron expression

timezone, enabled, lastRun, nextRun

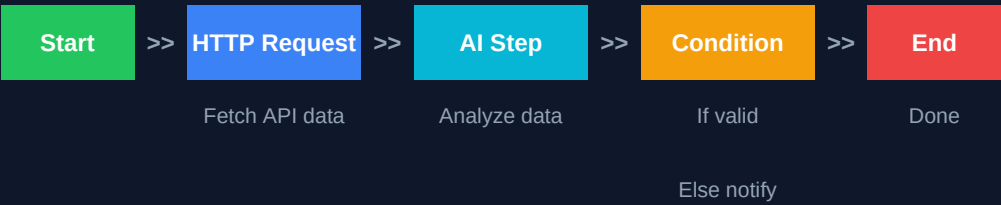
AuditLog

id, action, resource, resourceId

details (JSON), userId, ip, createdAt

Workflow Example

Automated Data Pipeline



Use Cases:

- Automated data extraction and transformation pipelines
- AI-powered content generation and review workflows
- Multi-step approval and notification systems
- Scheduled reporting and analytics automation
- Event-driven microservice orchestration

Deployment

GitHub Actions CI/CD Pipeline:

STAGE	ACTION	STATUS
Build	npm ci && npm run build	Automated
Test	TypeScript type checking	Automated
Deploy	GitHub Pages (frontend)	Automated
Backend	Vercel / Railway / Render	Manual

Deployment Options:

GitHub Pages - Static frontend hosting (free)

Vercel - Serverless backend deployment

Railway - Full-stack hosting with database

Render - Free tier for hobby projects

DigitalOcean App Platform - Scalable cloud hosting

Roadmap

Phase 1 - Core (Current)

Visual workflow editor with drag-and-drop

13 node types for building workflows

Multi-trigger support (Manual, Cron, Webhook)

REST API with full CRUD operations

Real-time execution monitoring

Phase 2 - Enhancement

User authentication and role-based access

Workflow versioning and rollback

Advanced error handling and retry logic

Workflow templates marketplace

Webhook management dashboard

Phase 3 - Intelligence

AI-powered workflow optimization

Natural language workflow creation

Predictive failure detection

Auto-scaling execution engine

Enterprise SSO integration

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Thank You

Intelligent Workflow Automation Agent

GitHub

github.com/habiutomo/intelligent-workflow-automation-agent

Live Demo

habiutomo.github.io/intelligent-workflow-automation-agent

Questions?

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